2. BitLocker

BitLocker is Microsoft's full-disk encryption feature that protects data if a computer is lost or stolen.

Purpose

- Encrypt the entire drive
- Prevent unauthorized access
- Protect sensitive business data
- Help meet security compliance requirements

How BitLocker Works

1. The drive is encrypted.
2. A **Trusted Platform Module (TPM)** or startup PIN/key verifies the device during boot.
3. If verification succeeds, Windows starts normally.
4. If not, BitLocker requests the recovery key.

BitLocker Recovery Key

A **48-digit recovery key** is used to unlock the drive when:

- Hardware changes occur
- TPM issues arise
- BIOS/UEFI settings change
- Incorrect PIN is entered repeatedly

Benefits

- Full-disk encryption
- Transparent user experience
- Protects data at rest
- Supports TPM security